Susan D. Suarez*
Gordon G. Gallup, Jr.

Department of Psychology, State University of New York at Albany, Albany, New York 12222, U.S.A.

Received 10 August 1980 and accepted 4 December 1980

Keywords: chimpanzees, orangutans, gorillas, mirrors, self-awareness.

Self-recognition in Chimpanzees and Orangutans, but not Gorillas

Recognition of one's own reflection in a mirror qualifies as an objective test of self-awareness. Although most primates appear incapable of learning that their behavior is the source of the behavior depicted in a mirror, the present study replicates previous reports showing that both chimpanzees and orangutans are capable of self-recognition. As the only remaining species of great ape, gorillas were also systematically tested with mirrors. Using a specially designed control procedure which provides independent evidence of interest in and motivation to touch unobtrusively applied marks used to assess self-recognition, gorillas appeared unable to correctly decipher mirrored information about themselves.

1. Introduction

Throughout history mankind has considered his species unique among all others, with special psychological endowments that place him at the pinnacle of creation (see Gallup & Suarez, in press). Recent findings with other primates, however, have begun to undermine many facets of this assumption (see Mason, 1976; Gallup *et al.*, 1977; Desmond, 1979). Self-awareness or self-consciousness, which can be defined by the capacity to become the object of one's own attention, is a trait believed in the past to be uniquely human, and, furthermore, unamenable to scientific investigation in other species (see Gallup, 1977, 1980). However, mirrors provide a persuasive means of testing the capacity for self-awareness in non-humans. When confronted with its reflection in a mirror an organism's ability to correctly identify the reflection presupposes a sense of self. If you do not know who you are, how could you possibly know who it is you are seeing when looking at yourself in a mirror? Without a sense of self, the source and significance of the reflection would forever remain unknown.

When presented with a mirror most species react as if they were seeing an unfamiliar conspecific and perform a variety of species-typical social responses such as threat and appeasement gestures directed at the reflection (Gallup, 1968). Many early investigators (e.g. Schmidt, 1878; Garner, 1914; Zeh, 1915; Yerkes, 1927) noticed that great apes, too, show an other-directed or social orientation when given brief access to a mirror. However, unlike other species, if the period of mirror exposure is extended chimpanzees soon learn to identify the source of activity in the reflection as being isomorphic with, and therefore a consequence of, their own behavior, and begin employing the mirror as a means of self-inspection and manipulation (i.e. self-directed behaviors). As an explicit test of self-recognition, Gallup (1970) anesthetized chimpanzees with extended prior exposure to mirrors and painted them with an odorless dye that left no tactile sensation on body parts that were visually inaccessible without the use of a mirror. Thus, since the chimpanzees were unconscious and therefore unaware of the marking procedure, could not feel or smell the dye, and could not see marked areas without a mirror, any behaviors directed toward touching and inspecting the marks when allowed to see a mirror would be evidence for

* Research conducted while a visiting Predoctoral Fellow at Yerkes Regional Primate Research Center, Emory University, Atlanta, Georgia 30322, U.S.A.

Journal of Human Evolution (1981) **10,** 175–188

0047–2484/81/020175+14 $02.00/0 © 1981 Academic Press Inc. (London) Limited

self-recognition. Upon reintroduction of the mirrors the chimpanzees repeatedly touched the marks, often followed by visual and olfactory inspection of the fingers which had touched the mark even though the dye was indelible. These results with chimpanzees have since been replicated a number of times (Hill *et al.*, 1970; Gallup *et al.*, 1971; Lethmate & Dücker, 1973), and the same findings have been reported for orangutans (Lethmate & Dücker, 1973). However, numerous attempts to assess self-recognition in a variety of "lesser" apes and monkeys have invariably yielded negative results (see Gallup *et al.*, 1980). These findings have been interpreted as indicating that such primates lack a cognitive category for processing mirrored information about the self, a deficit most likely tied to the absence of a sufficiently well-integrated concept of self (Gallup, 1977). It, therefore, appears that among primates self-recognition may be a trait unique to man and the rest of the great apes. In an attempt to further substantiate the accuracy of this statement the studies reported here were designed to assess the capacity for self-recognition in gorillas, the only remaining species of great ape for which no published data on self-awareness exists.

2. Gorillas

As in the original chimpanzee self-recognition study (Gallup, 1970), gorillas were given prolonged exposure to a mirror and monitored daily for behaviors directed at the reflection. As an objective test of self-recognition they were then anesthetized, marked with dye, and observed for their reactions to the marked areas when allowed to view their own reflections.

Subjects

One male and three female lowland gorillas (*Gorilla g. gorilla*) maintained at the Yerkes Regional Primate Research Center served as subjects. The male, Ozoum, was wild-born, estimated to be 19 years old, and periodically housed with females but at the time of the experiment lived alone. Two of the female subjects, Jini and Katoomba, were also wild-born and were housed together. Their ages were estimated to be 17 and 18 years, respectively. The remaining female, Inaki, was 13 years old, born in captivity, and housed with another female not used in the present study.

Procedure

Each gorilla was individually confined to the indoor section (2·4 × 2·4 × 2·0 m high) of its indoor-outdoor compound for 5 h per day and given exposure to a full-length Plexiglas mirror (76 × 152 cm for the male; 71 × 125 cm for females) situated approximately 0·5 m away from the cage door. The experimenter observed each subject for 15 min twice daily while quietly seated on a chair next to the mirror. All gorillas had previously been accustomed to having observers seated outside of the cage and did not appear to be affected by the experimenter's presence. Records were kept of the amount of time spent viewing the mirror, the number and description of social responses directed at the reflection (e.g. chest beat, charge, retreat), and the incidence of any behaviors directed at the self with the aid of the mirror. It was originally intended that the gorillas be given 10 days of mirror exposure in accord with the procedure used by Gallup (1970) for chimpanzees. However, unlike chimpanzees, no gorilla exhibited self-directed behavior in front of the mirror during this time and it was therefore decided that exposure

should be extended for 16 days. On the 17th day all gorillas were anesthetized with a 10 mg/kg dose of Ketamine HCl. Each subject was then marked over the entire brow ridge and across the left wrist with a pink dye produced by mixing white clown paint with red grease paint (M. Stein Cosmetics, New York). Two of the gorillas (Ozoum and Katoomba) were also marked over the left half of the top of the head with the white paint. All marked areas were lightly dusted with corn starch which we previously determined was effective in reducing any olfactory and/or tactile cues associated with the marks when applied to a human hand. Although it would have been preferable to mark the gorillas with a red dye so as to more closely replicate chimpanzee self-recognition studies (e.g. Gallup, 1970), their dark skin color would have provided a poor contrasting surface. Whereas red was chosen by Gallup because it simulates the appearance of blood and should command interest on biological grounds, pink was used here rather than some other light color on the assumption that it, too, should have some biological relevance since it simulates the color of genital openings and the inside of the mouth.

After complete recovery from anesthetization (4–5 h later) each gorilla was observed for 30 min to determine the number of times each marked area was touched. Once this baseline had been established a mirror was returned to the front of the cage. The experimenter observed the subject for an additional 30 min and recorded time spent viewing the reflection, the incidence of social responses and self-directed responses, and the number of times any marked regions were touched both with and without the use of the mirror.

Results

Figure 1 shows average viewing time per day, and Figure 2 shows the data for total number of social responses per day combined for all gorillas. Decreases across days in both time spent viewing the reflection and social behaviors directed at the mirror are evident. However it should be noted that, in contrast to chimpanzees (e.g. Gallup, 1970), responses directed at the self with the aid of the mirror were never observed in any of the gorillas. Nor was there an increase in viewing time on the test day as compared to preceding days. Responses directed at marked areas combined for all subjects are shown in Figure 3. All gorillas showed a keen interest in their painted wrists both before and after the reintroduction of the mirror. However, there were no instances in which a subject touched a marked area visually accessible only with the aid of the mirror while watching the reflection.

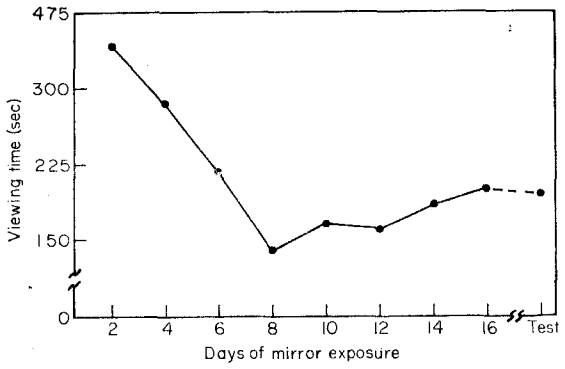


Figure 1. Mean time spent viewing the reflection by gorillas during recording sessions as a function of days of exposure.

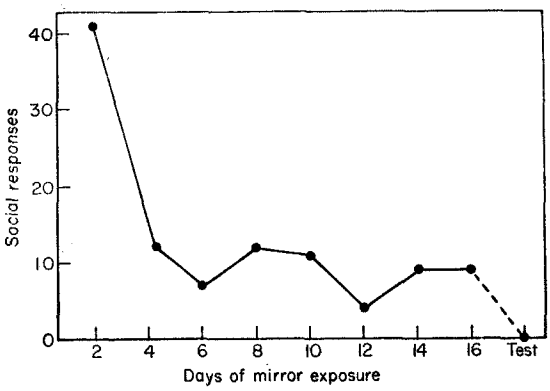


Figure 2. Total number of social responses by gorillas to their mirror image as a function of days of exposure.

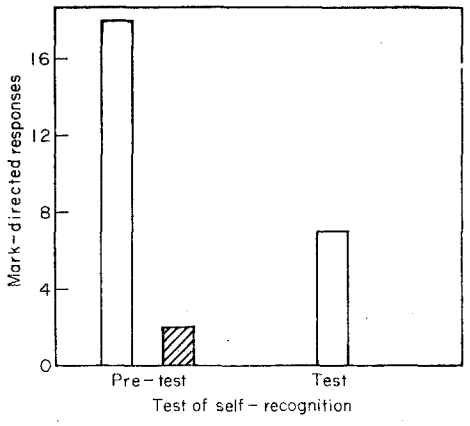


Figure 3. Total number of responses directed to marked eyebrows (▨) and wrists (□) by gorillas prior to mirror exposure and upon reintroduction of the mirror.

3. Orangutans

In an attempt to determine whether some situational factor peculiar to the Yerkes Primate Research Center might have been responsible for the gorillas' failure to evidence self-recognition, it was decided that orangutans, which have been reported capable of recognizing themselves (Lethmate & Dücker, 1973), should be tested under the same conditions as the gorillas.

Subjects

Subjects were two male orangutans (*Pongo pygmaeus*) maintained at the Yerkes Primate Research Center. Dyak was wild-born with an estimated age of 22 years and housed alone. Kanting was 12 years old, born at the center and raised in the nursery, and at the time of the experiment lived with another male not used in this study.

Procedure

The orangutans were confined to indoor cages identical to those of the gorillas and given 5 h of daily exposure to a Plexiglas mirror (76 × 152 cm for Dyak; 71 × 125 cm for Kanting) situated the same distance from the cage door. The same method of data

collection was used as for the gorillas with the exception that, due to the recurrent observation of behaviors directed at the self with the use of the mirror, only eight days of mirror exposure were given prior to the mark test. On the test day the orangutans were anesthetized with a 10 mg/kg dose of Ketamine HCl and marked with pink over the brow ridge and on the left wrist. Viewing time, social responses, and self-directed behaviors were recorded on the mark test using the same procedure used in the gorilla study.

Results

Viewing time and social responses for Kanting are plotted across days in Figures 4 and 5, respectively. In contrast to Kanting, the gorillas in our first study, and chimpanzees in previous studies Dyak was atypical in that he was never observed making any social responses to his mirror image. Furthermore, although his viewing time was comparable to that for Kanting prior to the mark test, Dyak did not show an increase in time spent watching the mirror on the test day as found with Kanting. Figure 6 shows Kanting's data for self-directed behaviors. Dyak also differed from Kanting in that whereas Kanting often performed self-directed behaviors in front of the mirror (e.g. inspection and grooming of the inside of the mouth, nose and eyes), Dyak was never observed using the mirror to gain information about parts visually inaccessible without a mirror. The incidence of

Figure 4. Mean time spent viewing the reflection by an orangutan during recording sessions as a function of days of exposure.

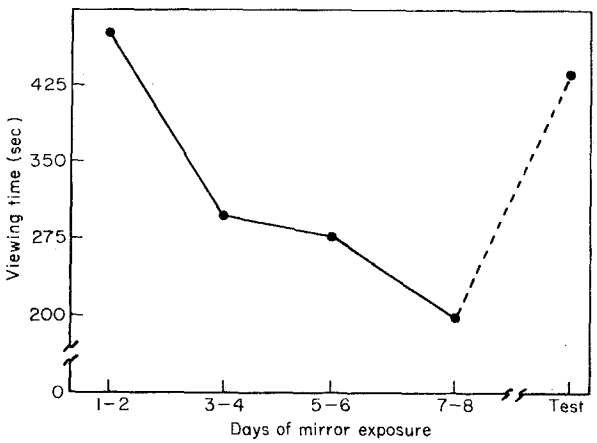
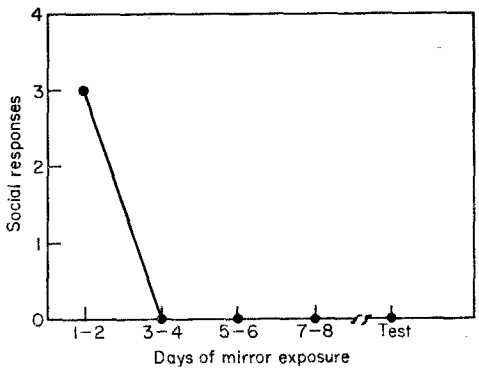


Figure 5. Number of social responses by an orangutan to its mirror image as a function of days of exposure.



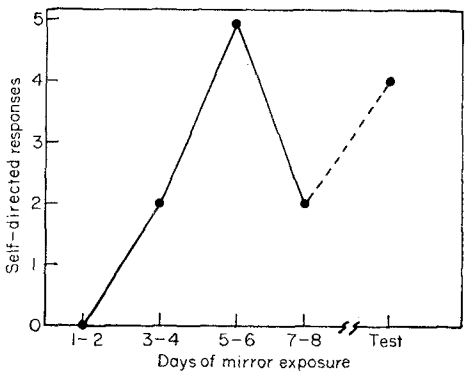


Figure 6. Number of self-directed responses by an orangutan in the presence of a mirror as a function of days of exposure.

mark-directed responses performed by Kanting is shown in Figure 7. As explicit evidence for his ability to recognize himself, Kanting showed an increase in the number of times he touched his marked brow after being allowed access to the mirror, and once smelled his finger after touching his brow. Dyak made no such attempts to touch or inspect his dyed brow. In contrast to the gorillas, whose failure to show mark-directed behaviors on the test could not be attributed to a lack of motivation to touch the marks as evidenced by their high rate of grooming a marked area they could see without a mirror, Dyak did not inspect his marked wrist. The results for his test were, therefore, rendered inconclusive.

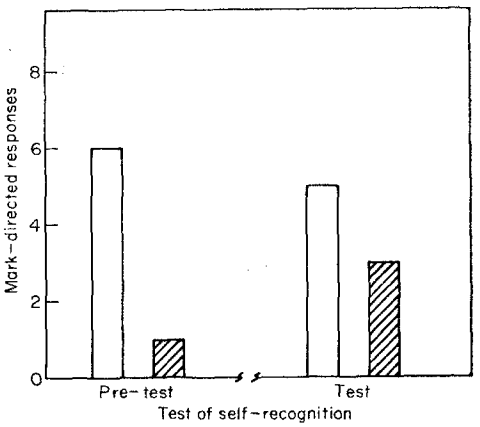


Figure 7. Number of responses directed to a marked eyebrow(▨) and wrist (□) by an orangutan prior to mirror exposure and upon reintroduction of the mirror.

4. Chimpanzees

In order to further verify that our failure to find self-recognition in gorillas was not due to some situational and/or procedural peculiarity, chimpanzees were also tested under the same conditions as the gorillas.

Subjects

One male and one female wild-born chimpanzee (*Pan troglodytes*) from the Yerkes Primate Research Center served as subjects. The male, Homer, was estimated to be 21 years of age and the female, Tai, was approximately 16 years of age. Both subjects were housed with opposite sexed chimpanzees not used in this study.

Procedure

The procedure used was identical to that described in the preceding study. After eight days of exposure to a mirror (76 × 152 cm for Homer; 71 × 125 cm for Tai) the chimpanzees were anesthetized with 10 mg/kg of Ketamine HCl and marked over the brow ridge and on the left wrist with the pink dye. Viewing times, social behaviors, self-directed behaviors and mark-directed responses were recorded as before.

Results

Average time spent viewing the reflection across days is plotted in Figure 8. Figures 9 and 10, respectively, depict mean number of social responses directed at the mirror and self-directed behaviors across days combined for both subjects. In support of previous findings (e.g. Gallup, 1970), the chimpanzees showed progressive decreases in viewing time and social responses across days, with a concomitant increase in the number of behaviors directed at themselves with the aid of the mirror. Some of the most noteworthy instances of self-directed behavior included daily vaginal swelling inspections by the female, sometimes while standing on her head in front of the mirror.

Figure 11 shows the number of mark-directed responses combined for both chimpanzees. Prior to being allowed to look at their reflections after anesthetization, both chimpanzees avidly groomed their painted wrists. Once the mirror was reintroduced attention rapidly

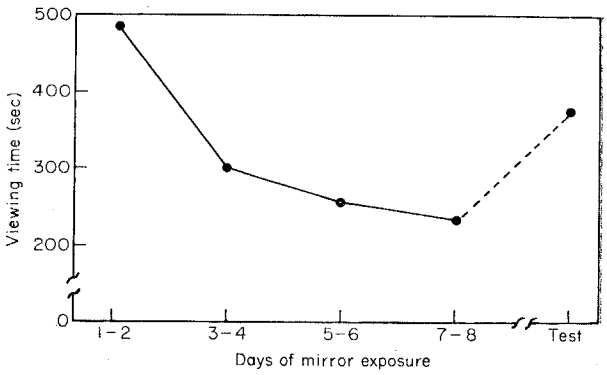


Figure 8. Mean time spent viewing the reflection by chimpanzees during recording sessions as a function of days of exposure.

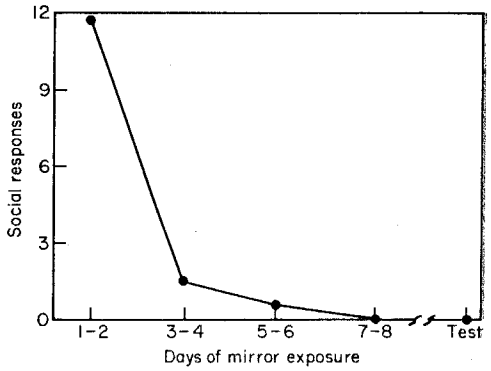


Figure 9. Mean number of social responses by chimpanzees to their mirror image as a function of days of exposure.

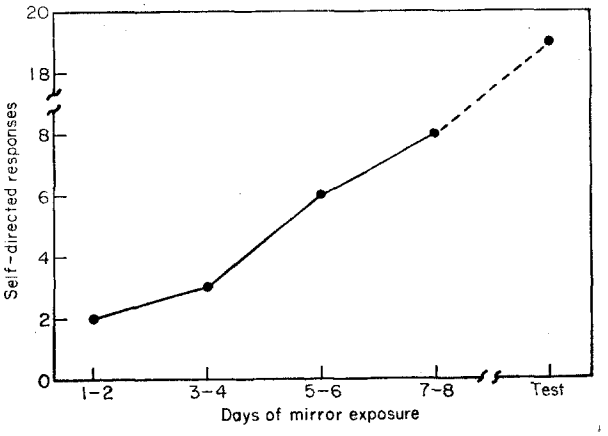


Figure 10. Total number of self-directed responses by chimpanzees in the presence of a mirror as a function of days of exposure.

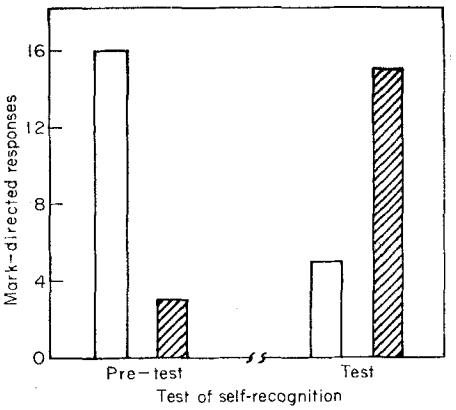


Figure 11. Total number of responses directed to a marked eyebrow (▨) and wrist (□) by chimpanzees prior to mirror exposure and upon reintroduction of the mirror.

shifted to the marked brow ridge. Both subjects were frequently observed not only touching their eyebrows but also attempting to scratch the mark off with a fingernail while gazing intently at the mirror. These behaviors were often followed by smelling and/or licking the fingers which had touched the mark.

5. Additional Study

Because it is generally assumed that the ability to recognize oneself in a mirror is a learned phenomenon arising, in part, from prolonged exposure to one's own reflection, earlier studies of self-recognition (e.g. Gallup, 1970) employed 10 days of mirror exposure prior to the mark test to allow ample time for learning. Yet self-directed behaviors typically emerge in chimpanzees during the first few days of mirror confrontation. The results of the preceding study show that eight days of exposure is sufficient to obtain positive results. Therefore, in the present study chimpanzees were given only four days of experience with their reflection in an effort to determine whether mirror exposure extended beyond the emergence of a self-directed orientation is necessary to obtain evidence of self-recognition on the mark test.

Subjects

Subjects were two male chimpanzees (*Pan troglodytes*) maintained at the Yerkes Primate Research Center. Hoboh was wild-born, estimated to be 20 years old, and lived with the female who participated in the preceding study. Nine-year old William was born at the Center and raised in the nursery, and presently housed with three juvenile chimpanzees of mixed sex.

Procedure

Housing conditions, data collection methods, and procedure for anesthetization and marking of the brow ridge and wrist were identical to those used in the preceding studies with the exception that the chimpanzees were given only four days of mirror exposure prior to the mark test.

Results

Average daily viewing time and total number of social and self-directed responses combined for both chimpanzees are shown in Table 1. Decreases in viewing time and social responses and an increase in self-directed behavior were observed in both subjects over days as in the preceding experiment. Mark-directed responses are given in Table 2.

Table 1 **Responses to a mirror by chimpanzees given four days of exposure**

	Day				
	1	2	3	4	Test
Average viewing time (sec)	547	147	194	93	246
Total social responses	22	0	0	0	0
Total self-directed behaviours	0	1	5	2	3

Table 2 **Mark-directed responses by chimpanzees given four days of exposure**

	Pre-test		Test	
	Wrist	Brow	Wrist	Brow
Hoboh	5	1	2	6
William	4	2	1	0

Both chimpanzees appeared highly interested in their marked wrists during the pre-test phase, but only Hoboh touched his dyed brow ridge (often followed by olfactory inspection of the investigating finger) when allowed access to the mirror. William's failure to show brow-directed responses is likely due to the fact that immediately prior to his return after the marking procedure his cage had been inadvertently hosed down. During recovery from anesthetization William spent a considerable amount of time resting his forehead against one of the wet cage bars, which resulted in the removal of a majority of the dye on his brow.

6. Discussion

In the present series of experiments we found that whereas chimpanzees and an orangutan showed clear evidence of self-recognition, gorillas did not. In fact, unlike the other great apes the behavior of gorillas in response to their own reflection closely resembled that reported for preadolescent monkeys (see Gallup *et al.*, 1980). For chimpanzees and the orangutan the decline in social or other-directed behavior to the mirror was invariably replaced by gradually emerging patterns of self-directed behavior, where subjects were observed to begin using the mirror to inspect and manipulate parts of their bodies which they had not seen before. That is, rather than continuing to respond to the mirror as such, they began to use the mirror to respond to themselves. The gorillas, on the other hand, never showed self-directed behavior at any time throughout the experiment, and on the test of self-recognition failed to evidence any behavior directed to facial marks. As further evidence that the gorillas remained oblivious to the source and significance of what was depicted in the mirror, on the mark test of self-recognition they showed virtually no change in viewing time (see Figure 1). In contrast, the chimpanzees and the orangutan all showed substantial increases in time spent watching the reflection when they were allowed to examine themselves in the mirror following the application of pink dye to brow ridges (see Figures 4 and 9 and Table 1). This abrupt reinstatement of visual attention to mirrors by marking facial features replicates previous results with chimpanzees (Gallup, 1970; Gallup *et al.*, 1971), and provides further indirect evidence of self-recognition.

Several lines of evidence argue against the gorillas' failure being due to some procedural or situational factor. In all instances subjects were carefully chosen so as to preclude confounding due to possible maturational differences. The average age of the gorillas (16·7 years) was closely comparable to that of the chimpanzees (17·2 years) and the orangutans (17·0 years). Moreover, the gorillas, chimpanzees and orangutans were tested in exactly the same setting. It is doubtful, then, that variables that may have prevented the gorillas from adequately seeing their reflection (e.g. inadequate lighting, insufficient mirror size, or cage bars blocking the view) are responsible for their failure to show a self-directed orientation. Eye structure and visual characteristics are very similar within the great apes (Young, 1973), and an optical examination of the gorillas conducted by veterinarians during the anesthetization phase of this study revealed no structural or refractive anomalies. Furthermore, to counter the possibility that the first two gorillas tested performed no mark-directed responses with the use of the mirror because of some undetermined factor that prevented them from seeing their painted brows, mark size was greatly increased on the remaining gorillas to the extent that half of the head was covered with dye. This also proved ineffective.

Although in isolation these findings might be viewed as suspect, other investigators (D. Ledbetter, pers. comm.; J. Lethmate, pers. comm.) have also obtained negative results in studies of self-recognition in gorillas. They, too, failed to see the self- and/or mark-directed behaviors in their subjects that is characteristic of chimpanzees and orangutans. However, due to the absence of some means of establishing the gorillas' motivation to touch a dyed body part the results of the mark tests performed by others are rendered uninterpretable. One could always argue that gorillas simply do not care about such marks. To obviate this possibility we instituted a specially designed control procedure whereby in addition to marking the brow ridge, we painted a wrist on each subject.

Although the gorillas seemed very interested in and repeatedly groomed their wrists, which they could easily see without a mirror, they were completely oblivious to their own marked eyebrows when given access to the reflection. Lack of curiosity in and motivation to touch the marks, then, is not a viable alternative interpretation of the gorillas' failure to evidence self-recognition.

One might argue that the mark test for self-recognition may simply be an inappropriate test for gorillas as compared to chimpanzees and orangutans because of possible species differences in eye–hand coordination or some such combination of visual and motor skills that may leave gorillas at a disadvantage in performing self- and mark-directed behaviors. But an abundance of data suggests that this is not the case. Tool implementation, for example, appears to require abilities concordant with that demanded of subjects in a test of self-recognition. In both instances visual and motor performance must be successfully coordinated to produce the desired end result. Parker (1969, 1978) and Rensch & Dücker (1966) found no differences between the great apes in manipulatory ability or capacity to effectively use tools. Likewise, the ability to solve patterned string problems or manual detour problems appears to rely on analogous capacities. Yet gorillas do not differ from the other great apes on either of these tasks (Riesen *et al.*, 1953; Fischer & Kitchener, 1965; Davis *et al.*, 1974).

Still another argument that might be advanced is that even though gorillas may not differ from chimpanzees and orangutans in visual-manual coordination, an element of distraction peculiar to gorillas may have been produced by having the mirror outside of the cage and forcing subjects to view an image disrupted by cage bars. However, in a study of attentional processes which measured the disruptive influence of irrelevant foreground cues, Rumbaugh & McCormack (1969) found that gorillas were no more distracted than either chimpanzees or orangutans. In fact, in a more recent study Rumbaugh *et al.* (1973) conclude that gorillas are actually less distracted by such cues than either of the other two species of great apes.

Finally, an inadequate amount of mirror exposure prior to the mark test does not appear useful as an explanation of the gorillas' failure to show self-recognition. The performance of gorillas on a variety of traditional learning tasks is largely indistinguishable from that of chimpanzees and orangutans. Equivalent performance with the other great apes has been shown by gorillas in learning set formation (Rumbaugh & Rice, 1962; Rumbaugh & McCormack, 1967; Hodos, 1970; Rumbaugh & Gill, 1973), simple discrimination, discrimination-reversal, and oddity concept formation (Rumbaugh & McCormack, 1967), and spatial delayed response (Fischer & Kitchener, 1965). With the current controversy over the existence or non-existence of language abilities in non-humans aside, Patterson (1978*b*) claims American Sign Language learning by her gorilla to be at a level commensurate with that reported for chimpanzees (e.g. Gardner & Gardner, 1978). The comparable performance of gorillas in tasks used to assess memory processes (Robbins & Bush, 1973) also attests to continuity of mental functioning within the great apes. Yet as compared with 16 days of exposure for gorillas, we obtained evidence for self-recognition in chimpanzees and an orangutan given half of that amount of exposure, and in a chimpanzee given only four days of experience with a mirror. Furthermore, Ledbetter's two gorillas were given over 400 h of exposure to a mirror over a period of several months without showing any signs that they recognized their reflection (pers. comm.).

Contrary to the impression created by Patterson (1978*a*) that the gorilla "Koko" can recognize herself on the basis of making reference to her image in a mirror with

sign language, labeling the reflection cannot be taken as evidence of self-recognition nor can responses to marks which are applied in anything less than an unobtrusive manner (for details see Gallup, 1979*a*).

The ability to recognize one's own reflection may also be subject to rearing influences. Unlike those maintained in group cages, chimpanzees reared in social isolation fail to show evidence of self-recognition (Gallup *et al.*, 1971). Indeed, this may be why the orangutan, Dyak, who was housed individually failed to show signs of recognizing his own reflection. The gorillas, however, had all had extensive exposure to conspecifics yet failed to show self-recognition.

The absence of self-recognition in gorillas has some interesting implications for theories of self-awareness. For instance, many investigators contend that certain of Piaget's stages of sensori-motor development are involved in self-awareness. The attainment of object permanence has, in particular, been singled out as a necessary prerequisite to the development of self-recognition [for a review see Gallup (1979*a*)]. However, Redshaw (1978) reports that gorilla and human infants under 18 months of age perform similarly as far as "sensori-motor intelligence" is concerned, and pass through the same stages and sequence in developing a concept of object permanence. Our failure to find self-recognition in gorillas suggests, then, that although an understanding of object permanence and related concepts may be necessary, they clearly are not sufficient for self-recognition.

Taxonomic classifications can be based on genetic distance as well as morphology. It is well established that chimpanzees, gorillas and orangutans are not only genetically very closely related to each other but also to humans (e.g. Bruce & Ayala, 1978; Mitchell & Gosden, 1978). One researcher, in fact, contends on this basis that gorillas are more closely related to humans than either of the other great apes (Miller, 1977). If, however, taxonomic classification were to be based on behavioral traits that differentiate species to the extent that the capacity for self-recognition does, inclusion of gorillas as members of the Pongids might have to be qualified. On the other hand, it is important to acknowledge that individuals within certain categories of *Homo sapiens* (e.g. mentally retarded, schizophrenic) sometimes also lack the capacity to recognize themselves in mirrors (see Gallup, 1979*a*).

In conclusion, it is difficult to explain the lack of evidence for self-recognition in gorillas when they are in a variety of other respects so much like the other great apes. To date, no theories have been advanced specifying a neurological basis for self-conscious experience. However, since behavior in the last analysis is nothing more than an expression of neuronal activity, it seems reasonable that species differences in the capacity for self-awareness are ultimately reducible to neurological differences. While the answer does not appear to be tied to differences in lateralization since self-recognition is not restricted to either hemisphere in humans (Sperry *et al.*, 1979), it is possible that gorillas and other species which show no evidence of self-recognition lack some as yet unspecified neuroanatomical features.

In the event that a concept of self can be present but in a subthreshold form that does not readily lend itself to inquiry in the usual tests of self-recognition, we are currently conducting an extensive series of studies with monkeys in an attempt to elicit self-directed behavior through the use of procedures designed to make the identity of the mirror image more explicit (e.g. Gallup *et al.*, 1980). If success is obtained with these techniques suggestions for alternative means of assessing self-recognition in gorillas would be forth-

coming. A definitive answer to this issue is important because of the implications self-awareness has for intellectual functioning. To be able to conceive of oneself opens up a vast array of new conceptual vistas, including the potential to begin contemplating one's own individual demise, the development of an explicit species identity, and even the capacity to conceive of evolution (Gallup, 1979*b*).

This research was supported by Yerkes Center base grant No. RR-00165 and a National Science Foundation Pre-doctoral Fellowship held by the first author. We gratefully acknowledge the help of the Yerkes Primate Research Center director, Dr F. A. King and Dr R. D. Nadler. We especially wish to thank Dr King for the suggestion to conduct the second and third experiments. Thanks are also due to the Yerkes veterinarians for performing the anesthetizations, and the many animal caretakers whose assistance was vital to this project.

References

Bruce, E. J. & Ayala, F. J. (1978). Humans and apes are genetically very similar. *Nature* **276,** 264–265.

Davis, R. T., Medin, D. L. & Borkhuis, M. L. (1974). *Monkeys as Perceivers.* New York: Academic Press.

Desmond, A. J. (1979). *The Ape's Reflexion.* New York: Dial Press.

Fischer, G. J. & Kitchener, S. L. (1965). Comparative learning in young gorillas and orangutans. *Journal of Genetic Psychology* **107,** 337–348.

Gallup, G. G., Jr. (1968). Mirror-image stimulation. *Psychological Bulletin* **70,** 782–793.

Gallup, G. G., Jr. (1970). Chimpanzees: self-recognition. *Science* **167,** 86–87.

Gallup, G. G., Jr. (1977). Self-recognition in primates: a comparative approach to the bidirectional properties of consciousness. *American Psychologist* **32,** 329–338.

Gallup, G. G., Jr. (1979*a*). Self-recognition in chimpanzees and man: a developmental and comparative perspective. In (M. Lewis & L. A. Rosenblum, Eds) *Genesis of Behavior, Volume 2: The Child and Its Family.* New York: Plenum Press.

Gallup, G. G., Jr. (1979*b*). Self-awareness in primates. *American Scientist* **67,** 417–421.

Gallup, G. G., Jr. (1980). Chimpanzees and self-awareness. In (M. A. Roy, Ed.) *Species Identity and Attachment: A Phylogenetic Evaluation.* New York: Garland STPM Press.

Gallup, G. G., Jr., Boren, J. L., Gagliardi, G. J. & Wallnau, L. B. (1977). A mirror for the mind of man, or will the chimpanzee create an identity crisis for *Homo sapiens*? *Journal of Human Evolution* **6,** 303–313.

Gallup, G. G., Jr., McClure, M. K., Hill, S. D. & Bundy, R. A. (1971). Capacity for self-recognition in differentially reared chimpanzees. *Psychological Record* **21,** 69–74

Gallup, G. G., Jr. & Suarez, S. D. (in press). Overcoming our resistance to animal research: Man in comparative perspective. In (D. W. Rajecki, Ed.) *Comparing Behavior: Studying Man Studying Animals.* New Jersey: Lawrence Erlbaum Associates.

Gallup, G. G., Jr., Wallnau, L. B. & Suarez, S. D. (1980). Failure to find self-recognition in mother–infant and infant–infant rhesus monkey pairs. *Folia Primatologica* **33,** 210–219.

Gardner, R. A. & Gardner, B. T. (1978). Comparative psychology and language acquisition. *Annals of the New York Academy of Sciences* **309,** 37–76.

Garner, R. L. (1914). Gorillas in their own jungle. *Bulletin of the New York Zoological Society* **17,** 1062–1066.

Hill, S. D., Bundy, R. A., Gallup, G. G., Jr. & McClure, M. K. (1970). Responsiveness of young nursery-reared chimpanzees to mirrors. *Proceedings of the Louisiana Academy of Sciences* **33,** 77–82.

Hodos, W. (1970). Evolutionary interpretation of neutral and behavioral studies of living vertebrates. In (F. O. Schmidt, Ed.) *The Neurosciences: Second Study Program.* New York: Rockefeller University Press.

Lethmate, J. & Dücker, G. (1973). Untersuchungen zum Selbsterkennen in Spiegel bei Orang-utans und einigen anderen affenarten. *Zeitschrift fuer Tierpsychologie* **33,** 248–269.

Mason, W. A. (1976). Environmental models and mental modes: representational processes in the great apes and man. *American Psychologist* **31,** 284–294.

Miller, D. A. (1977). Evolution of primate chromosomes. *Science* **198,** 1116–1124.

Mitchell, A. R. & Gosden, J. R. (1978). Evolutionary relationships between man and the great apes. *Scientific Progress, Oxford* **65,** 273–293.

Parker, C. E. (1969). Responsiveness, manipulation, and implementation behavior in chimpanzees, gorillas, and orangutans. In *Proceedings of the Second International Congress on Primatology*. New York: Basel & Karger.

Parker, C. E. (1978). Opportunism and the rise of intelligence. *Journal of Human Evolution* **7,** 597–608.

Patterson, F. (1978*a*). Conversations with a gorilla. *National Geographic* **154,** 438–465.

Patterson, F. G. (1978*b*). The gestures of a gorilla: Language acquisition in another pongid. *Brain and Language* **5,** 72–97.

Redshaw, M. (1978). Cognitive development in human and gorilla infants. *Journal of Human Evolution* **7,** 133–141.

Rensch, B. & Dücker, G. (1966). Manipulatory ability of a young orangutan and a young gorilla. *Zeitschrift fuer Tierpsychologie* **23,** 874–892.

Riesen, A. H., Greenberg, B., Granston, A. S. & Fantz, R. L. (1953). Solutions of patterned string problems by young gorillas. *Journal of Comparative and Physiological Psychology* **46,** 19–22.

Robbins, D. & Bush, C. T. (1973). Memory in great apes. *Journal of Experimental Psychology* **97,** 344–348.

Rumbaugh, D. M. & Gill, T. V. (1973). The learning skills of great apes. *Journal of Human Evolution* **2,** 171–179.

Rumbaugh, D. M., Gill, T. V. & Wright, S. C. (1973). Readiness to attend to visual foreground cues. *Journal of Human Evolution* **2,** 181–188.

Rumbaugh, D. M. & McCormack, C. (1967). The learning skills of primates: a comparative study of apes and monkeys. In (D. Stark, R. Schneider & H. J. Kuhn, Eds) *Progress in Primatology*. Stuttgart: Fischer.

Rumbaugh, D. M. & McCormack, C. (1969). Attentional skills of great apes compared with those of gibbons and squirrel monkeys. In *Proceedings of the Second International Congress on Primatology*. New York: Basel & Karger.

Rumbaugh, D. M. & Rice, C. P. (1962). Learning-set formation in young great apes. *Journal of Comparative and Physiological Psychology* **55,** 866–868.

Schmidt, M. (1878). Beobachtungen am Orang-Utan. Cited in Yerkes, R. M. & Yerkes, A. W. (1929). *The Great Apes*. New Haven: Yale University Press.

Sperry, R. W., Zaidel, E. & Zaidel, D. (1979). Self-recognition and social awareness in the deconnected minor hemisphere. *Neuropsychologia* **17,** 153–166.

Yerkes, R. M. (1927). The mind of a gorilla. *Genetic Psychology Monographs* **2,** 1–193.

Young, F. A. (1973). Visual refractive characteristics and the subhuman primate. In (G. H. Bourne, Ed.) *Non-human Primates in Medical Research*. New York: Academic Press.

Zeh, L. E. (1915). A rare baby gorilla. Cited in Yerkes, R. M. & Yerkes, A. W. (1929). *The Great Apes*. New Haven: Yale University Press.